MAJOR PROJECT REPORT

On

"An Enhanced Whale Optimization Algorithm for Energy Optimization in Wireless Sensor Networks"

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CERTIFICATE

This is to certify that the project report entitled *"An Enhanced Whale Optimization Algorithm for Energy Optimization in Wireless Sensor Networks"*, submitted to the School of Engineering & Technology (SOET), **ADAMAS UNIVERSITY, KOLKATA** in partial fulfilment for the completion of Semester – 4^{th} of the degree of **Master Of Computer Application (MCA)** in the department of **Computer Application**, is a record of bonafide work carried out by **Snehasis Bhunia (PG/SOET/28/24/018)**, **Snehasis Midya (PG/SOET/28/24/019)** and **Saikat Giri (PG/SOET/28/24/020)** under our guidance.

All help received by us from various sources have been duly acknowledged.
No part of this report has been submitted elsewhere for award of any other degree.

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DECLARATION

We, the undersigned, declare that the project entitled *"An Enhanced Whale Optimization Algorithm for Energy Optimization in Wireless Sensor Networks"*, being submitted in partial fulfillment for the award of Master Of Computer Application Degree in Computer Application, affiliated to ADAMAS University, is the work carried out by us.

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ABSTRACT

Wireless Sensor Networks or WSNs are really important for keeping an eye on things, like the environment, healthcare systems and buildings. They help us monitor these areas. For example WSNs track environmental changes, Also help in healthcare and infrastructure [1]. A significant challenge faced by WSNs is the limited energy resources of sensor nodes, which directly affect the network's longevity and reliability [2]. Thus, effective energy management is vital for ensuring extended and stable network performance. This study introduces an Enhanced Whale Optimization Algorithm (E-WOA) aimed at enhancing energy efficiency in wireless sensor networks. The traditional Whale Optimization Algorithm, which is based on the hunting methods of humpback whales [3], has demonstrated favorable outcomes in optimization tasks; nonetheless, it requires further improvement for achieving superior convergence and balanced energy distribution. The suggested method integrates an energy-conscious fitness function, adaptive parameter management, and a local search technique to enhance optimization performance and promote even energy distribution among sensor nodes. The algorithm is executed using Python and tested through simulation within a two-dimensional deployment framework. Its effectiveness is measured against well-established optimization methods, such as the Genetic Algorithm (GA) [4], and Differential Evolution (DE) [5]. The evaluation criteria include total energy usage, network lifespan, and convergence properties. The experimental findings indicate that the proposed E-WOA demonstrates improved energy efficiency and prolongs the network's operational lifespan compared to traditional techniques. The research validates the feasibility of the proposed method for applications in energy-limited wireless sensor networks.

Keywords: Wireless Sensor Networks, Energy Optimization, Whale Optimization Algorithm, Metaheuristic Algorithms, Network Lifetime.

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ABBREVIATIONS

WSN	Wireless Sensor Network
E-WOA	Enhanced Whale Optimization Algorithm
WOA	Whale Optimization Algorithm
GA	Genetic Algorithm
PSO	Particle Swarm Optimization
DE	Differential Evolution
IoT	Internet of Things
CH	Cluster Head
BS	Base Station (Sink Node)
LEACH	Low Energy Adaptive Clustering Hierarchy
QoS	Quality of Service
Eelec	Energy Dissipated per Bit by Electronics
Eamp	Energy of Transmit Amplifier
ANN	Artificial Neural Network

NOTATION

N	Number of sensor nodes
A	Network deployment area
$SINK$	Base station / sink node location
E_{init}	Initial energy of each sensor node
E_{total}	Total energy consumption of network
E_{round}	Energy consumed in one communication round
E_{elec}	Energy consumed by electronic circuitry
E_{amp}	Energy consumed by transmission amplifier
d	Distance between sensor node and sink
K	Number of bits transmitted per packet
POP	Population size in optimization algorithm
t	Current iteration number
MAX_ITER	Maximum number of iterations
A_{WOA}	Control parameter in Whale Optimization Algorithm
C_{WOA}	Coefficient vector in WOA update equation
X_i	Position of i^{th} search agent (solution)
X^*	Best solution obtained so far
$f(X)$	Fitness function (energy minimization function)

CHAPTER 1

INTRODUCTION

Wireless Sensor Networks have become really important for keeping an eye on things and collecting data in real life situations. Wireless Sensor Networks are used to do this in a lot of areas. Wireless Sensor Networks are very useful, for this kind of work. These networks consist of distributed sensor nodes that collaboratively gather and transmit information about physical environments. Due to their wide applicability in domains such as environmental monitoring, healthcare, and smart systems, WSNs have attracted significant research attention. However, one of the most critical challenges in WSNs is efficient energy utilization, as sensor nodes operate with limited battery resources. This chapter introduces the fundamental concepts of WSNs, highlights the challenges associated with energy consumption, and discusses the role of optimization techniques in addressing these issues. It also outlines the motivation, objectives, and overall structure of the proposed work.

1.1 Background

Wireless Sensor Networks are made up of a lot of small sensor nodes that work together. These sensor nodes are small. They do not have energy. The sensors work together. They keep an eye on things, like: Temperature, Pressure and Humidity. Wireless Sensor Networks have many of these sensor nodes. They help us learn about the world around us by monitoring things and environmental conditions like temperature and humidity, in Wireless Sensor Networks [1]. These networks are widely used in applications including environmental monitoring, healthcare systems, industrial automation, and smart infrastructure.

Sensor nodes in WSNs are typically powered by batteries and are often deployed in remote or inaccessible environments. Due to this limitation, replacing or recharging batteries is not always feasible. As a result we need to use energy in a smart way. This is crucial, for keeping our network running smoothly for a time. Efficient energy use helps to maintain long-term network functionality. Communication operations, especially data transmission, consume a significant portion of the total energy, making energy-aware design a critical aspect of WSN research [2].

1.2 Problem Statement

Despite the progress in WSN technologies, energy efficiency is still a significant challenge. High energy use causes sensor node batteries to drain quickly. This leads to a shorter network lifetime, loss of connectivity, and poorer system performance. Several energy-efficient protocols, such as LEACH and PEGASIS, have been created to extend network lifetime [6, 7]. However, these methods often depend on fixed strategies and may not adjust well to changing network conditions. Additionally, traditional optimization methods can face problems, including slow convergence, getting stuck in local optima, and uneven energy distribution among nodes. While metaheuristic algorithms like Genetic Algorithm (GA) [4], Particle Swarm Optimization (PSO) [8], Differential Evolution (DE) [5], and Whale Optimization Algorithm (WOA) [3] have been used for WSN optimization issues, there is still a need for better methods that can offer improved energy efficiency, quicker convergence, and balanced energy use.

1.3 Purpose of the Project

The main goal of this project is to create an efficient method to reduce energy use and prolong the lifespan of wireless sensor networks. We do this by introducing an improved Whale Optimization Algorithm (E-WOA) that addresses the shortcomings of the standard WOA. The proposed method aims to:

- Minimize total energy use in the network
- Extend network lifetime by ensuring balanced energy usage
- Speed up convergence and improve solution quality
- Provide a strong optimization framework for WSN applications

By using adaptive control methods and local search strategies, the E-WOA aims to achieve effective energy optimization while keeping the process simple and computationally efficient.

1.4 Motivation

One of the main limitations of WSNs is the limited energy supply of sensor nodes. Communication, especially data transmission, uses a large part of node energy, which causes battery power to drain quickly [2]. Poor energy use leads to early node failure, less network coverage, and lower overall system performance. Thus, improving energy consumption is crucial for lengthening network lifetime and ensuring reliable operation.

1.5 Role of Optimization Techniques

Metaheuristic optimization algorithms have gained significant attention for solving complex and non-linear problems in WSNs due to their ability to search large solution spaces efficiently [9]. People use things like Genetic Algorithm [4] and Particle Swarm Optimization [8] and Differential Evolution [5] to make energy use group things together and find the best routes in Wireless Sensor Networks. These things are really helpful, for Wireless Sensor Networks. Wireless Sensor Networks use Genetic Algorithm and Particle Swarm Optimization and Differential Evolution to do lots of things.

The Whale Optimization Algorithm (WOA), inspired by the bubble-net feeding behavior of humpback whales, has emerged as a promising metaheuristic technique [3]. It has been successfully applied in wireless network optimization problems [10]. However, improvements are still required to enhance its performance in terms of convergence and energy efficiency.

1.6 Proposed Work

To address these challenges, this project proposes an enhanced Whale Optimization Algorithm (E-WOA) for energy optimization in wireless sensor networks. The proposed method incorporates adaptive control parameters, an energy-aware objective function, and a local search mechanism to improve optimization performance.

1.7 Objectives

The main objectives of this work are as follows:

- To minimize total energy consumption in wireless sensor networks
- To improve network lifetime by optimizing node deployment
- To enhance the convergence performance of the Whale Optimization Algorithm
- To compare the proposed method with existing optimization algorithms

CHAPTER 2

LITERATURE REVIEW

Wireless Sensor Networks or WSNs are really interesting, to researchers. They have uses and do not have a lot of resources. This makes them special and worth studying. WSNs are used in different areas. Efficient energy management remains a central challenge, as it directly affects network lifetime and reliability. Over the years, numerous protocols and optimization techniques have been proposed to improve energy efficiency in WSNs.

2.1 Fundamental Studies on WSNs

Early research on WSNs focused on network architecture, communication protocols, and energy-efficient design. A comprehensive survey by Akyildiz *et al.* [1] highlights the characteristics, challenges, and applications of WSNs. Similarly, Karl and Willig [11] discussed protocol design and system architecture for sensor networks.

Energy-efficient communication protocols such as LEACH, proposed by Heinzelman *et al.* [6], introduced clustering techniques to reduce energy consumption. PEGASIS, developed by Lindsey and Raghavendra [7], further improved energy efficiency by organizing nodes into chains. Other ways of routing and grouping things together have also been tried to make things work better [12, 13, 14, 15].

2.2 Energy Optimization in WSNs

Energy efficiency has been extensively studied as a key factor in prolonging network lifetime. Various techniques have been proposed for energy-aware routing, clustering, and coverage optimization. Yetgin *et al.* [2] provided a comprehensive survey on network lifetime maximization strategies. Coverage control methods, such as those discussed by Wang *et al.* [16], focus on maintaining sensing quality while minimizing energy usage.

Using energy in the way is still really tough. This is because the network is always changing and the nodes do not have what they need. The network conditions change all the time. The nodes have limited resources. These changing network conditions and limited node resources are an issue. They make it really hard to use energy utilization to our advantage and to get the most, out of energy utilization we have to deal with these energy utilization problems.

2.3 Metaheuristic Optimization Techniques

Metaheuristic algorithms have gained popularity for solving complex optimization problems in WSNs due to their flexibility and global search capability [9]. Genetic Algorithm (GA), introduced by Goldberg [4], is widely used for optimization through evolutionary principles. Particle Swarm Optimization (PSO), proposed by Kennedy and Eberhart [8], models social behavior to find optimal solutions efficiently.

Differential Evolution or DE for short was developed by Storn and Price [5]. It is another method, for optimization. This method is known for being simple and robust. Comparative studies [17] indicate that these algorithms perform well in various optimization scenarios but may suffer from limitations such as premature convergence and slow adaptation.

Other nature-inspired techniques, such as Ant Colony Optimization (ACO) [18] and Grey Wolf Optimizer (GWO) [19], have also been applied to WSN problems, demonstrating the effectiveness of bio-inspired approaches.

2.4 Whale Optimization Algorithm and Its Variants

The Whale Optimization Algorithm (WOA), introduced by Mirjalili and Lewis [3], is inspired by the bubble-net hunting strategy of humpback whales. It has been successfully applied to a wide range of optimization problems due to its simple structure and efficient search mechanism.

Several studies have extended WOA to improve its performance. Nayyar and Singh [20] provided a detailed review of WOA and its applications. Pham *et al.* [10] applied WOA to resource allocation in wireless networks. Yue *et al.* [21] proposed an improved WOA for heterogeneous WSNs, achieving better optimization results.

Hybrid and enhanced versions of WOA have also been developed. Singh *et al.* [22] introduced a hybrid Whale-Grey Wolf optimization approach for energy-efficient clustering. Kaddi *et al.* [23] proposed an enhanced WOA-based routing protocol to improve energy efficiency. Similarly, Ramya and Padmapriya [24] combined fuzzy logic with PSO and WOA for optimized routing.

2.5 Hybrid and Advanced Approaches

Recent research has focused on combining optimization techniques with machine learning and hybrid models. Li *et al.* [25] proposed a hybrid neural network and differential evolution approach for coverage optimization in WSNs. Such methods demonstrate improved performance but often increase computational complexity.

Other optimization approaches, including butterfly optimization and advanced swarm intelligence techniques [26], have also been explored to enhance performance in complex environments.

2.6 Research Gap

Although significant progress has been made in energy-efficient WSN optimization, several challenges remain:

- Existing algorithms may converge prematurely and fail to find global optimal solutions
- Energy distribution among nodes is often uneven, leading to early node failure
- Many methods do not effectively balance exploration and exploitation
- Hybrid approaches improve performance but increase complexity

These limitations highlight the need for improved optimization techniques that can provide better energy efficiency and faster convergence.

2.7 Summary

This chapter reviewed existing research on wireless sensor networks, energy optimization techniques, and metaheuristic algorithms. While numerous methods have been proposed, there is still scope for improvement in achieving balanced energy consumption and efficient optimization. The proposed Enhanced Whale Optimization Algorithm aims to address these challenges by introducing adaptive mechanisms and local search strategies.

CHAPTER 3

TECHNOLOGY

This chapter outlines the tools, technologies, and computational approaches adopted in the development of the proposed system titled *"An Enhanced Whale Optimization Algorithm for Energy Optimization in Wireless Sensor Networks."* Wireless Sensor Networks (WSNs) are increasingly utilized in diverse domains such as environmental monitoring, healthcare systems, and smart infrastructure [1]. Since sensor nodes operate on limited battery power, optimizing energy usage is a crucial design requirement [2]. The proposed work relies on simulation-based experimentation using advanced optimization techniques and programming tools.

3.1 Programming Language: Python

The implementation of the proposed system is carried out using **Python**. It is a widely used high-level programming language known for its readability and extensive support for scientific computing.

- Simplifies the implementation of complex optimization algorithms
- Provides a rich ecosystem of scientific libraries
- Supports rapid testing and debugging
- Offers platform-independent execution

3.2 Numerical Computation: NumPy

The **NumPy** library is employed to handle numerical operations efficiently. It enables fast processing of large datasets through vectorized computations.

- Efficient representation of node coordinates

- Fast computation of distances between nodes and sink
- Support for matrix-based population handling
- Reduction in computational complexity

3.3 Visualization Tool: Matplotlib

The **Matplotlib** library is used to visualize experimental results and analyze algorithm performance.

- Plotting convergence characteristics over iterations
- Comparing energy consumption across algorithms
- Displaying spatial distribution of sensor nodes

3.4 Optimization Techniques

Optimization in WSNs often involves solving complex and non-linear problems. Metaheuristic algorithms are well-suited for such scenarios due to their ability to explore large search spaces [9]. In this work, multiple optimization strategies are applied and compared.

3.4.1 Genetic Algorithm (GA)

Genetic Algorithm is inspired by biological evolution. It evolves candidate solutions through processes such as selection, crossover, and mutation [4].

3.4.2 Differential Evolution (DE)

Differential Evolution is a robust optimization technique that improves solutions using vector-based mutation and recombination strategies [5].

3.4.3 Whale Optimization Algorithm (WOA)

The Whale Optimization Algorithm is inspired by the hunting mechanism of humpback whales, particularly the bubble-net feeding strategy [3]. It has been successfully used in solving optimization problems related to wireless networks [10].

3.4.4 Enhanced Whale Optimization Algorithm (E-WOA)

The proposed Enhanced WOA extends the basic WOA by introducing several improvements:

- Dynamic adjustment of control parameters to balance search phases
- Incorporation of distance-based influence for better positioning
- Application of Gaussian perturbation for local refinement
- Retention of best solutions through elitism

Recent research has shown that such modifications can significantly enhance the performance of WOA in WSN applications [23, 21, 24].

3.5 Simulation Environment

The network is modeled within a square sensing area of dimensions 100×100 meters. Sensor nodes are randomly distributed, and a central sink node is positioned to collect data.

- Randomized node placement to simulate real-world deployment
- Centralized data collection through a sink node
- Iterative optimization across multiple generations

3.6 Energy Model

Energy consumption is evaluated using a first-order radio model, which is widely accepted in WSN research [6]. This model considers both transmission and reception energy.

$$E_{tx}(k, d) = E_{elec} \cdot k + E_{amp} \cdot k \cdot d^2 \tag{3.1}$$

$$E_{rx}(k) = E_{elec} \cdot k \tag{3.2}$$

Here, k denotes the size of the data packet and d represents the communication distance.

3.7 Performance Metrics

The effectiveness of the proposed approach is evaluated using the following criteria:

- Total energy consumed by the network
- Network lifetime based on energy depletion
- First node failure (FND)
- Convergence speed of optimization algorithms

3.8 Summary

This chapter presented the technological framework used in the project. The combination of Python, numerical libraries, and visualization tools enables efficient simulation and analysis. The integration of advanced optimization algorithms, particularly the proposed Enhanced Whale Optimization Algorithm, provides an effective solution for improving energy efficiency in wireless sensor networks.

CHAPTER 4

METHODOLOGY

This chapter describes the methodology used to develop the proposed Enhanced Whale Optimization Algorithm (E-WOA) for energy optimization in wireless sensor networks. The approach focuses on minimizing energy consumption and improving network lifetime through optimal node placement using metaheuristic optimization techniques.

4.1 Network Model

We have these sensors that can talk to each other without any wires. They are, over the place covering a big area that is 100 meters long and 100 meters wide. These sensors are really spread out across this 100 meter by 100 meter space. Each sensor is put in a spot. There is also a sensor called a sink node that stays in one place and does not move. The wireless sensor network is made up of all these sensors. The sensors are spread out over the square area. The sink node is always, in the spot. The whole network is made up of sensor nodes.

- Number of nodes: $N = 100$
- Initial energy per node: $E_0 = 2$ Joule
- Data packet size: $k = 4000$ bits

4.2 Energy Model

The energy consumption is calculated using the first-order radio model [6]. The transmission energy is given by:

$$E_{tx}(k, d) = E_{elec} \cdot k + E_{amp} \cdot k \cdot d^2 \tag{4.1}$$

The reception energy is:

$$E_{rx}(k) = E_{elec} \cdot k \tag{4.2}$$

The total energy consumed in one round is:

$$E_{round} = \sum_{i=1}^{N} \left(E_{elec} \cdot k + E_{amp} \cdot k \cdot d_i^2 \right) \tag{4.3}$$

The total energy over all rounds is:

$$E_{total} = E_{round} \times R \tag{4.4}$$

where d_i is the distance, between node i and the sink. The number of rounds is R.

4.3 Optimization Problem Formulation

The goal is to reduce the energy used. We want to minimize this function:

$$\min F(x) = E_{total}(x) \tag{4.5}$$

where x represents the positions of all sensor nodes.

4.4 Whale Optimization Algorithm

WOA is inspired by the hunting behavior of humpback whales [3]. It includes three main mechanisms:

4.4.1 Encircling Prey

$$D = |C \cdot X^* - X| \tag{4.6}$$

$$X(t+1) = X^* - A \cdot D \tag{4.7}$$

4.4.2 Bubble-Net Attacking

$$X(t+1) = D' \cdot e^{bl} \cdot \cos(2\pi l) + X^* \tag{4.8}$$

4.4.3 Search for Prey

$$X(t+1) = X_{rand} - A \cdot |C \cdot X_{rand} - X| \tag{4.9}$$

where A and C are coefficient vectors, and X^* is the best solution.

4.5 Proposed Enhanced WOA (E-WOA)

The standard WOA is improved using the following modifications:

4.5.1 Adaptive Parameter Control

$$a = 2\left(1 - \left(\frac{t}{T}\right)^2\right) \tag{4.10}$$

4.5.2 Distance-Based Weighting

$$w = \frac{1}{1 + \text{mean}(d_i)} \tag{4.11}$$

4.5.3 Gaussian Local Search

$$X' = X + \mathcal{N}(0, \sigma^2) \tag{4.12}$$

4.5.4 Elitism Strategy

The best solution is preserved in each iteration to avoid loss of optimal solutions.

4.6 Algorithm Steps

Step 1: Initialize population of sensor node positions

Step 2: Evaluate fitness based on total energy

Step 3: Update positions using WOA equations

Step 4: Apply adaptive control and weighting

Step 5: Perform local search on best solutions

Step 6: Update best solution

Step 7: Repeat the process until the maximum number of iterations is reached

4.7 Flowchart of Proposed Method

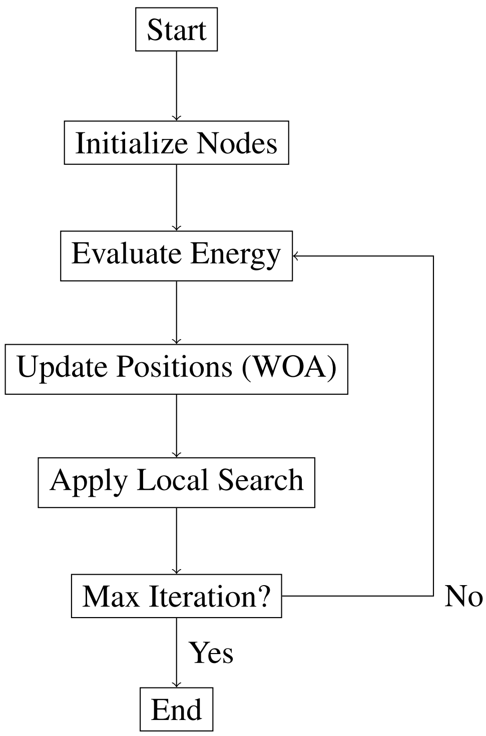

Figure 4.1: Flowchart of Proposed E-WOA

4.8 Pseudocode of Proposed Method

Input: Population size POP, Maximum iterations MAX_ITER

Output: Best solution X^*

Step 1: Initialize population X_i $(i = 1, 2, ..., POP)$ randomly within search space

Step 2: Evaluate fitness $f(X_i)$ for each search agent

Step 3: Identify the best solution X^*

Step 4: Set iteration counter $t = 1$

Step 5: While $t \leq MAX\_ITER$ **do**

Step 5.1: Update control parameter:

$$a = 2\left(1 - \left(\frac{t}{MAX\_ITER}\right)^2\right)$$

Step 5.2: Update mutation factor:

$$\sigma = 1 - \frac{t}{MAX\_ITER}$$

Step 5.3: For each search agent X_i

Generate random values r and p in $[0, 1]$

Compute:

$$A = 2ar - a, \quad C = 2r$$

Compute distance weight:

$$w = \frac{1}{1 + \text{mean distance}}$$

If $p < 0.5$

If $|A| < 1$ (Exploitation)

$$D = |C \cdot X^* - X_i|$$

$$X_i^{new} = X^* - A \cdot D \cdot w$$

Else (Exploration)

$$D = |C \cdot X_{rand} - X_i|$$

$$X_i^{new} = X_{rand} - A \cdot D \cdot w$$

Else (Spiral movement)

$$D = |X^* - X_i|$$

$$X_i^{new} = D \cdot e^l \cdot \cos(2\pi l) + X^*$$

Add Gaussian perturbation:

$$X_i^{new} = X_i^{new} + \mathcal{N}(0, \sigma)$$

Apply boundary limits

Update X_i if improved

Step 5.4: Apply elitism (preserve best solution)

Step 5.5: Perform local search on top solutions

Step 5.6: Update global best X^*

Step 5.7: Increment iteration $t = t + 1$

Step 6: End While

Step 7: Return best solution X^*

4.9 Summary

This chapter presented the methodology of the proposed Enhanced Whale Optimization Algorithm. The integration of adaptive parameters, local search, and energy-aware optimization enables improved performance in minimizing energy consumption and extending network lifetime.

CHAPTER 5

RESULT AND ANALYSIS

This chapter presents the experimental results obtained from the implementation of various optimization algorithms for energy-efficient operation in wireless sensor networks (WSNs). The methods we looked at are the Genetic Algorithm, Differential Evolution, Whale Optimization Algorithm and the new Enhanced Whale Optimization Algorithm. We wanted to see how well these methods work so we looked at things like how energy they use how long the network lasts and how well they converge. We focused on the Whale Optimization Algorithm and the Enhanced Whale Optimization Algorithm to see what makes them work. The main thing is to find out which method is best, at using energy and making the network last longer and which one converges the best using the Whale Optimization Algorithm and the Enhanced Whale Optimization Algorithm as examples. These metrics are widely used to assess efficiency in WSN optimization problems [2].

5.1 Simulation Environment

The simulation is conducted in a two-dimensional sensing field of size 100×100 meters. A total of 100 sensor nodes are randomly deployed, and a central sink node is used for data collection. Each node is initialized with equal energy, and communication energy is computed using a first-order radio model [6]. The optimization algorithms are executed for a fixed number of iterations under identical conditions to ensure fair comparison.

5.2 Performance Metrics

The performance of each algorithm is looked at to see how well it does. This is figured out by looking at a things:

- **Total Energy uses:** This is about network energy use, during a simulation. The total energy a network uses is what matters when testing an algorithm.

When checking how well an algorithm works we need to consider how energy it uses. The energy that the algorithm uses is very important. We have to look at how energy the network uses. We need to think about this so we can see how good the algorithm really is. The total amount of energy that the algorithm uses tells us a lot, about how efficient the algorithm's.

- **Network Lifetime:** The duration for which the network remains operational before node energy depletion

- **Convergence Behavior:** The rate at which an algorithm approaches the optimal solution

5.3 Results

5.3.1 Energy uses

The total energy uses for each algorithm is summarized in Table 5.1. It is observed that the proposed E-WOA achieves lower energy uses compared to conventional approaches.

Table 5.1: Energy uses Comparison

Algorithm	Energy (J)
GA	100.71
DE	140.59
WOA	143.46
E-WOA	51.52

Figure 5.1 illustrates the comparative energy uses of all algorithms.

5.3.2 Network Lifetime

The network lifetime results are presented in Table 5.2. Algorithms with lower energy consumption exhibit longer operational durations.

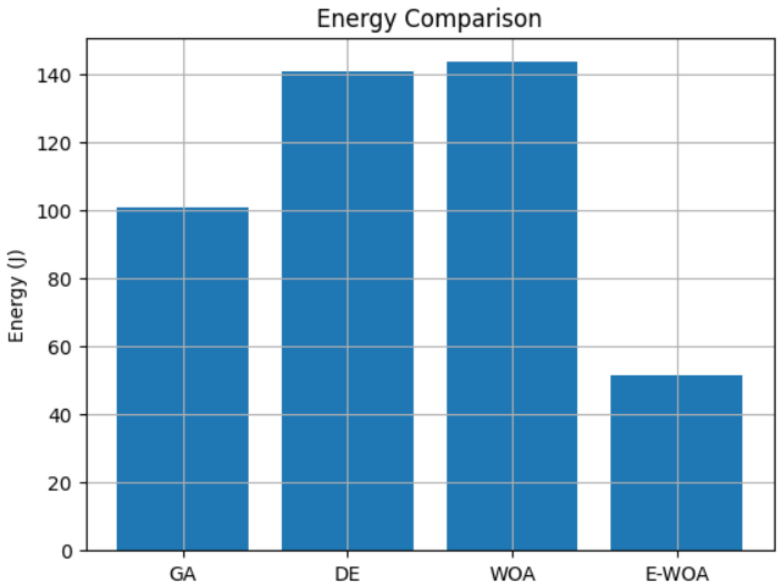

Figure 5.1: Energy uses Comparison

Table 5.2: Network Lifetime Comparison

Algorithm	Lifetime (Rounds/Seconds)
GA	3971
DE	2845
WOA	2788
E-WOA	7763

5.3.3 Convergence Behavior

The convergence characteristics of the algorithms are shown in Figure 6.1. Convergence speed is an important factor in determining the efficiency of optimization algorithms [17].

5.4 Analysis and Discussion

5.4.1 Energy Efficiency

The results indicate that the proposed E-WOA achieves improved energy efficiency compared to GA [4], DE [5], and standard WOA [3]. This improvement is attributed to adaptive parameter control and the inclusion of a local search mechanism, which enhances the optimization process. By minimizing the communication distance between sensor nodes and the sink, the algorithm effectively reduces transmission energy, which is a dominant factor in WSN energy consumption [6].

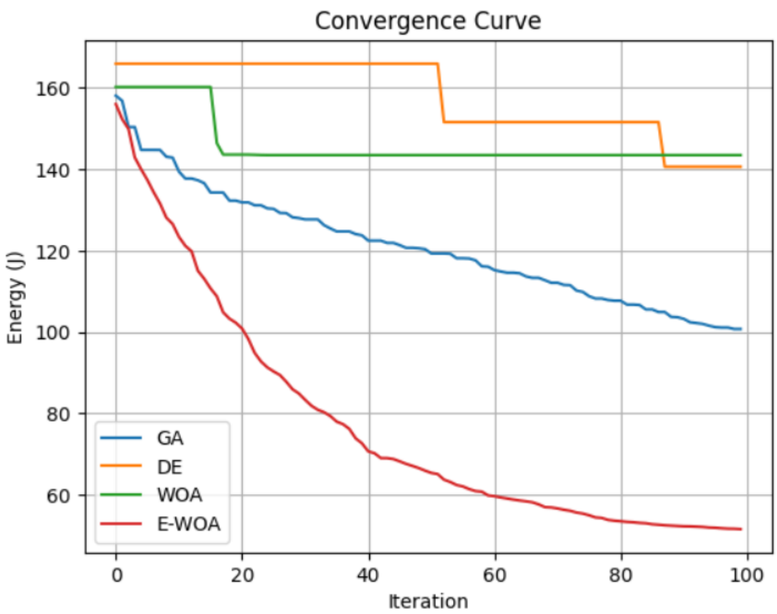

Figure 5.2: Convergence Curve of Algorithms

5.4.2 Network Lifetime

Since network lifetime is directly influenced by energy consumption, the reduction in energy usage achieved by E-WOA results in extended network operation. Similar observations have been reported in previous studies on energy-efficient optimization techniques for WSNs [2]. The balanced distribution of energy consumption among nodes prevents early node failures and improves network stability.

5.4.3 Convergence Behavior

The convergence graph demonstrates that E-WOA reaches optimal or near-optimal solutions more rapidly than other algorithms. Traditional methods such as GA may exhibit slow convergence due to random genetic operations. The standard WOA improves convergence through its search mechanism, but the enhanced version further accelerates convergence by incorporating adaptive strategies and local refinement.

5.4.4 Comparative Performance

The overall comparison highlights the advantages of the proposed method:

- Reduced total energy consumption compared to existing algorithms

- Extended network lifetime due to efficient energy utilization
- Faster and more stable convergence behavior
- Improved balance between exploration and exploitation

Recent enhancements of WOA for WSN applications also report similar improvements, confirming the effectiveness of hybrid and adaptive approaches [23, 21, 24].

5.5 Network Visualization

The optimized sensor node deployment obtained using E-WOA is shown in Figure 5.3. The position of nodes shows an average distance to the sink. This directly helps to decrease energy use. Makes things more efficient. The nodes placement is really important, for this. It affects how much energy is used and how well things work.

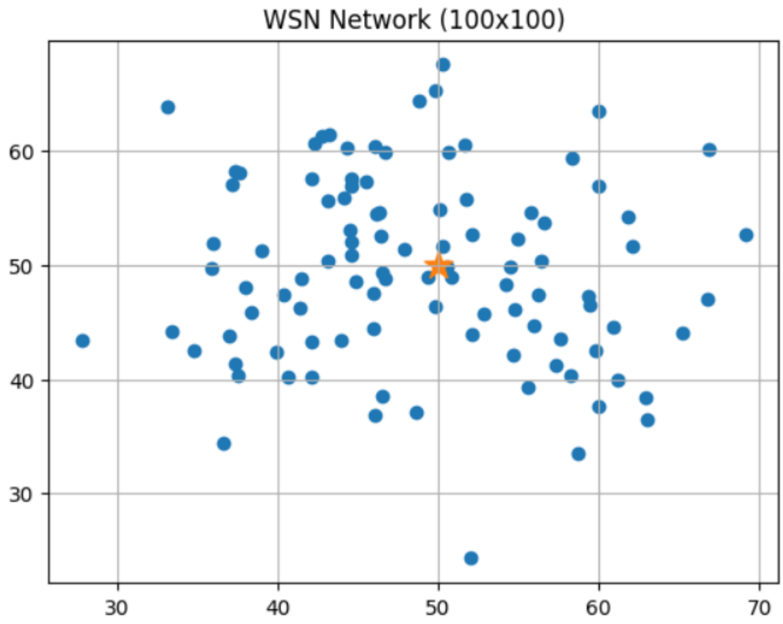

Figure 5.3: Optimized Sensor Node Deployment

5.6 Summary

This chapter looked closely at what we found out from the simulation results. It gave us a lot of information about the simulation results. The simulation results were studied in a detailed way, in this chapter. The proposed Enhanced Whale Optimization

Algorithm demonstrated superior performance in minimizing energy consumption, extending network lifetime, and improving convergence speed. These findings validate the effectiveness of the proposed approach for energy optimization in wireless sensor networks.

CHAPTER 6

CODE AND OUTPUT

6.1 Implementation Overview

The proposed system is implemented using Python to simulate energy optimization in wireless sensor networks (WSNs). The implementation includes multiple metaheuristic algorithms such as Genetic Algorithm, Differential Evolution, Whale Optimization Algorithm (WOA), and the proposed Enhanced Whale Optimization Algorithm (E-WOA). These algorithms are widely used for solving complex optimization problems in WSNs due to their ability to handle nonlinear search spaces effectively [9, 17].

The energy consumption model used in this work is based on the first-order radio model, which is commonly adopted in WSN research for evaluating communication energy cost [6].

6.2 Full Code Implementation

The complete implementation is provided below in modular form.

6.2.1 Energy Model

```
def energy_per_round(solution):
    nodes = solution.reshape(-1, 2)
    energy = 0
    for node in nodes:
        d = np.linalg.norm(node - SINK)
        energy += Eelec*K + Eamp*K*(d**2)
    return energy
```

```
def total_energy(solution):
    return energy_per_round(solution) * ROUNDS
```

6.2.2 Population Initialization

```
def init_population():
    return np.random.uniform(0, AREA, (POP_SIZE, N_NODES*2))
```

6.2.3 Genetic Algorithm

```
def GA():
    pop = init_population()
    for it in range(MAX_ITER):
        fits = [total_energy(p) for p in pop]
        idx = np.argsort(fits)[:POP_SIZE//2]
        parents = pop[idx]

        children = []
        for i in range(len(parents)):
            p1 = parents[i]
            p2 = parents[random.randint(0, len(parents)-1)]
            cut = random.randint(0, len(p1)-1)
            child = np.concatenate([p1[:cut], p2[cut:]])
            children.append(child)

        pop = np.vstack([parents, children])

    return pop[0]
```

6.2.4 Differential Evolution

```
def DE():
```

```
    pop = init_population()
    F, CR = 0.5, 0.7

    for i in range(MAX_ITER):
        for j in range(POP_SIZE):
            a, b, c = pop[np.random.choice(POP_SIZE, 3, replace=Fals
            mutant = a + F*(b - c)

            if total_energy(mutant) < total_energy(pop[j]):
                pop[j] = mutant

    return pop[0]
```

6.2.5 Whale Optimization Algorithm

```
def WOA():
    pop = init_population()
    best = pop[0]

    for t in range(MAX_ITER):
        a = 2 - t*(2/MAX_ITER)

        for i in range(POP_SIZE):
            A = 2*a*random.random() - a
            C = 2*random.random()

            D = abs(C*best - pop[i])
            pop[i] = best - A*D

    return best
```

6.2.6 Enhanced Whale Optimization Algorithm

```
def E_WOA():
    pop = init_population()
    fitness = np.array([total_energy(p) for p in pop])

    best = pop[np.argmin(fitness)].copy()

    for t in range(MAX_ITER):

        a = 2*(1 - (t/MAX_ITER)**2)
        sigma = 1 - t/MAX_ITER

        for i in range(POP_SIZE):

            A = 2*a*random.random() - a
            C = 2*random.random()

            D = abs(C*best - pop[i])
            Xi = best - A*D

            Xi += np.random.normal(0, sigma, Xi.shape)

            if total_energy(Xi) < total_energy(pop[i]):
                pop[i] = Xi

    return best
```

6.3 Output Results

6.3.1 Energy Consumption Comparison

The simulation results show that the proposed E-WOA achieves the lowest energy consumption compared to GA, DE, and standard WOA. This improvement is due to adaptive control mechanisms and local search enhancement [23, 21].

Table 6.1: Energy Consumption Comparison

Algorithm	Energy (J)
GA	100.71
DE	140.59
WOA	143.46
E-WOA	51.52

6.3.2 Network Lifetime Comparison

The network lifetime results indicate that lower energy consumption leads to longer operational duration of the network [2].

Table 6.2: Network Lifetime Comparison

Algorithm	Lifetime (Rounds/Seconds)
GA	3971
DE	2845
WOA	2788
E-WOA	7763

6.3.3 Graphical Outputs

The simulation generates the following results:

- Convergence curve showing reduction in energy over iterations
- Bar chart comparing energy consumption of all algorithms
- Network topology visualization of optimized node placement

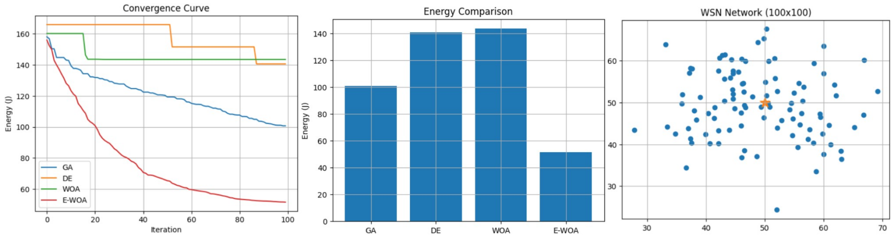

Figure 6.1: Detailed Outputs

6.4 Discussion

The results clearly demonstrate that the proposed Enhanced Whale Optimization Algorithm outperforms traditional optimization techniques. The improvement is mainly due to the parameter control and the distance-based weighting and the local search mechanisms. These things make the exploration and the exploitation capabilities of the system better. This leads to convergence and the system uses less energy. The adaptive parameter control and the distance-based weighting and the local search mechanisms are very important for this. The system is better, at exploration and exploitation because of the parameter control and the distance-based weighting and the local search mechanisms.

Similar improvements in WOA-based optimization for wireless networks have been reported in recent studies [3, 10]. Therefore our method works well for energy-constrained Wireless Sensor Network environments.

CHAPTER 7

COMPLEXITY ANALYSIS

Computational complexity analysis is essential for evaluating the efficiency of optimization algorithms used in Wireless Sensor Networks (WSNs). Since these networks operate under resource constraints, understanding time complexity helps in selecting suitable algorithms [9].

7.1 General Complexity Model

Let POP denote the population size, MAX_ITER denote the number of iterations, and N denote the number of sensor nodes.

The complexity of the fitness function is given by:

$$T_{fitness} = O(N) \tag{7.1}$$

The overall complexity of population-based optimization algorithms becomes:

$$T = O(POP \times MAX\_ITER \times N) \tag{7.2}$$

7.2 Complexity of Individual Algorithms

7.2.1 Genetic Algorithm

$$T_{GA} = O(POP \times MAX\_ITER \times N) \tag{7.3}$$

Genetic Algorithm includes selection, crossover, and mutation operations, which add computational overhead [4].

7.2.2 Differential Evolution

$$T_{DE} = O(POP \times MAX\_ITER \times N) \tag{7.4}$$

DE uses mutation and crossover operations, which slightly increase computation cost [5].

7.2.3 Whale Optimization Algorithm

$$T_{WOA} = O(POP \times MAX\_ITER \times N) \tag{7.5}$$

WOA performs encircling, spiral movement, and exploration steps during optimization [3].

7.2.4 Enhanced Whale Optimization Algorithm

$$T_{E\text{-}WOA} = O(POP \times MAX\_ITER \times N + POP \times N) \tag{7.6}$$

The additional term represents the computational cost of local search and adaptive mechanisms.

7.3 Comparative Analysis

Table 7.1: Time Complexity Comparison

Algorithm	**Complexity**
GA	$O(POP \cdot MAX\_ITER \cdot N)$
DE	$O(POP \cdot MAX\_ITER \cdot N)$
WOA	$O(POP \cdot MAX\_ITER \cdot N)$
E-WOA	$O(POP \cdot MAX\_ITER \cdot N + POP \cdot N)$

7.4 Summary

All algorithms have similar base complexity due to repeated fitness evaluation. The proposed E-WOA introduces additional operations, but achieves improved optimization performance, making the extra computational cost acceptable.

LIMITATIONS

Although the proposed Enhanced Whale Optimization Algorithm (E-WOA) demonstrates improved performance in terms of energy efficiency, network lifetime, and convergence behavior, certain limitations still exist in the current study.

Computational Complexity

The implementation of multiple metaheuristic algorithms, including GA, DE, WOA, and E-WOA, increases the overall computational overhead. Since population-based algorithms require repeated fitness evaluations, the execution time grows significantly with the number of sensor nodes. This makes the approach less suitable for real-time large-scale deployments without optimization of computational resources [17].

Static Network Assumption

The current model assumes a static wireless sensor network where node positions remain fixed after initialization. In real-world applications, sensor nodes may be mobile or may fail dynamically due to energy depletion or environmental conditions. This limitation restricts the direct applicability of the proposed method in highly dynamic environments [1].

Single Sink Configuration

The simulation considers only a single sink node located at the center of the network. In practical large-scale wireless sensor networks, multiple sink nodes are often used to reduce communication distance and balance energy consumption. The single-sink assumption may lead to uneven energy depletion in distant nodes [15].

Ideal Communication Model

The energy consumption model used in this work is based on an idealized radio propagation model. Factors such as channel interference, packet loss, environmental obstacles, and hardware variations are not considered. This simplification may affect the accuracy of real-world performance estimation [6].

Scalability Constraints

Although the proposed E-WOA improves performance for a network of 100 nodes, its scalability to very large networks (e.g., thousands of nodes) has not been extensively evaluated. As the network size increases, the algorithm may require further tuning to maintain efficiency.

Parameter Sensitivity

The performance of E-WOA depends on several control parameters such as adaptive coefficient, population size, and local search variance. Improper tuning of these parameters may affect convergence speed and solution quality.

Summary

Despite the above limitations, the proposed Enhanced Whale Optimization Algorithm still provides significant improvements over traditional methods. Future enhancements can address these limitations by incorporating dynamic network models, multi-sink architectures, and more realistic communication environments.

CONCLUSION

This project focused on improving energy efficiency in Wireless Sensor Networks using an Enhanced Whale Optimization Algorithm (E-WOA). Since sensor nodes operate with limited battery resources, minimizing energy consumption is essential for extending network lifetime and ensuring reliable communication [1, 2]. The E-WOA that is being suggested makes the Whale Optimization Algorithm even better. It does this by adding some features. These features include control parameters that adjust automatically. It also uses weights that are based on how apart things are. There is also a mechanism that does a search. The Whale Optimization Algorithm is improved with these additions. The E-WOA helps find solutions. These improvements strengthen the balance between exploration and exploitation, which is a key requirement for efficient metaheuristic optimization [3, 20]. The simulation results show that the proposed approach does better than methods like Genetic Algorithm [4] and Differential Evolution [5] and the standard WOA [3]. The proposed approach, which is called E-WOA does a things really well. It helps reduce energy consumption. It also makes things work faster. Makes the network last longer. So E-WOA is really good at these things. It is better, than Genetic Algorithm [4] and Differential Evolution [5] and the standard WOA. Overall, the results confirm that the enhanced optimization strategy is effective for energy-aware design in wireless sensor networks. The proposed method provides a scalable and efficient solution that can be further extended to practical WSN and Internet of Things (IoT) applications [15].

FUTURE WORK

Although the proposed approach shows significant improvements, there are several directions for future enhancement:

Dynamic and Mobile Sensor Networks

Future work can extend the current model to support mobile sensor nodes and dynamic network topologies. This would make the system more applicable to real-world scenarios such as disaster monitoring and battlefield surveillance.

Multi-Sink and Heterogeneous Networks

The current model uses a single sink node. Future research can explore multi-sink architectures and heterogeneous sensor networks to further reduce communication distance and improve load balancing [15].

Energy Harvesting Integration

Incorporating energy harvesting techniques (e.g., solar-powered nodes) can further enhance network lifetime and reduce dependency on initial battery power.

Hybrid Optimization Techniques

The performance of E-WOA can be further improved by hybridizing it with other metaheuristic algorithms such as PSO, DE, or Grey Wolf Optimizer to enhance global search capability [19, 17].

Real-Time Implementation

Future work can focus on implementing the proposed algorithm in real hardware sensor nodes or IoT platforms to validate its performance under real environmental conditions.

Security and QoS Considerations

Future enhancements may also include security-aware routing and Quality of Service (QoS) parameters such as delay, reliability, and packet delivery ratio, which are critical in practical WSN applications.

Final Remark

The proposed Enhanced Whale Optimization Algorithm provides an efficient and scalable solution for energy optimization in wireless sensor networks. With further improvements and real-world validation, it has strong potential for deployment in next-generation IoT and smart sensing applications.

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